

Web Application Security Testing using Kali Linux

Prepared by: Sabhya Kumari

Role:- Cybersecurity Analyst Intern

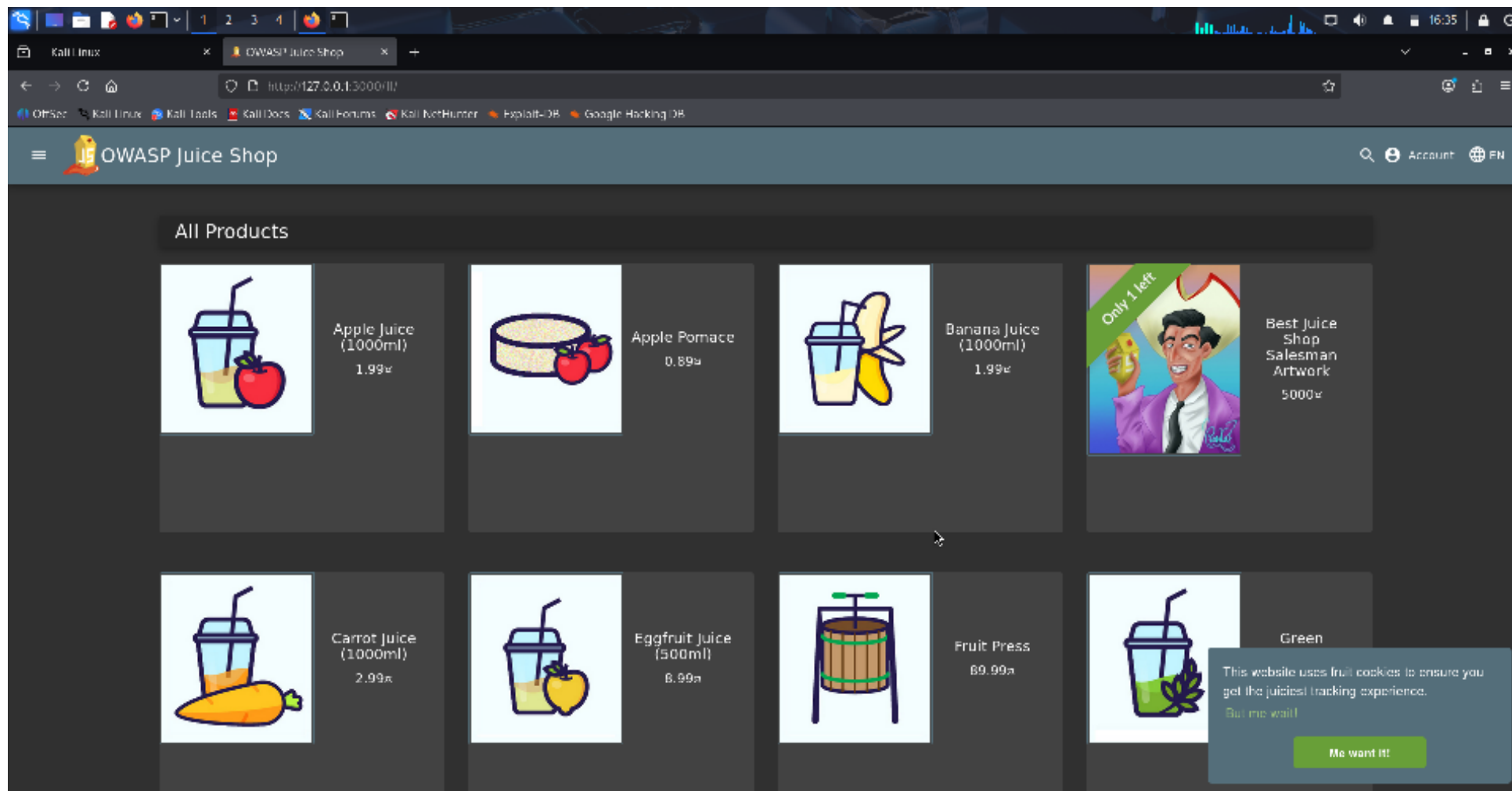
Date:- 29-04-2026

Objective:

- To To understand basic web security testing
- To analyze network and application traffic
- To identify vulnerabilities

Tools	Details
Attacker OS	Kali Linux (VMware)
Target App	OWASP Juice Shop
RAM Allocated	4 GB
CPUs	4
Network	NAT(Internet) + Host-Only(Lab)

●Juice-Shop Homepage

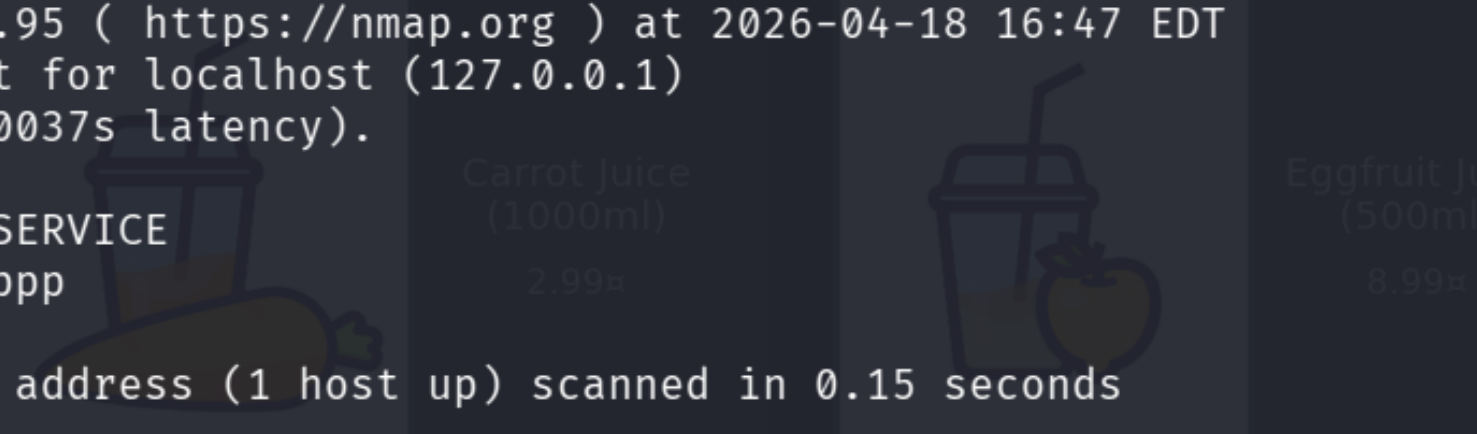


● Docker PS(Juice-Shop Running)

```
(kali㉿kali)-[~] - java (1.80_3) ...  
$ sudo docker ps  
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS                               NAMES  
6e2599abde48   bkimminich/juice-shop   "/nodejs/bin/node /j..." 12 seconds ago   Up 11 seconds   0.0.0.0:3000→3000/tcp, [::]:3000→3000/tcp   juice-shop
```

● Nmap Scan(Port 3000 Open)

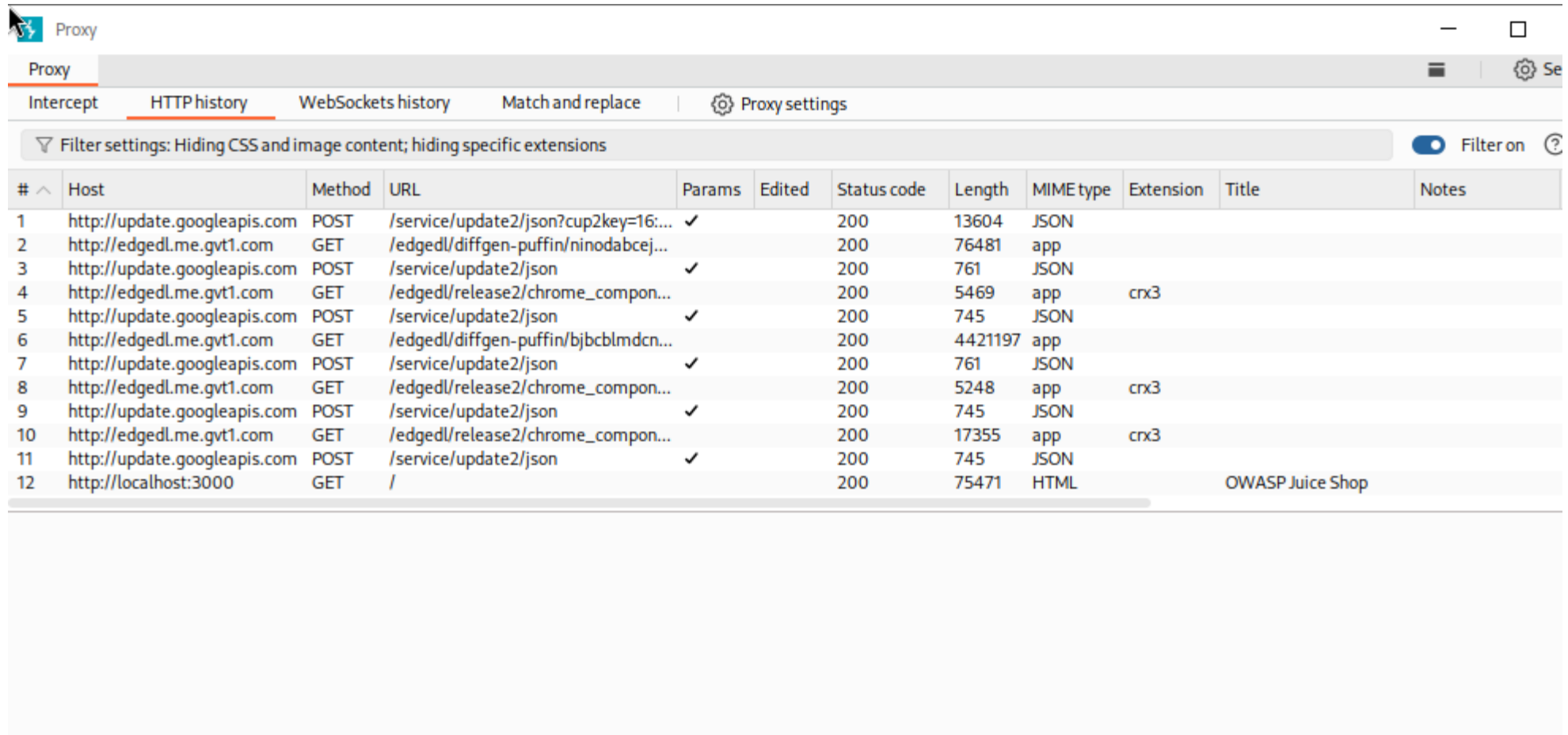
```
(kali㉿kali)-[~]  
$ nmap -p 3000 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-18 16:47 EDT  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.00037s latency).  
  
PORT      STATE SERVICE  
3000/tcp  open  ppp  
  
Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
```



●IP Configuration(Kali Linux)

```
(kali㉿kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:74:cf:dd brd ff:ff:ff:ff:ff:ff  
    inet 192.168.60.128/24 brd 192.168.60.255 scope global dynamic noprefixroute eth0  
        valid_lft 1747sec preferred_lft 1747sec  
    inet6 fe80::3ab6:112d:4852:6538/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:74:cf:e7 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.191.128/24 brd 192.168.191.255 scope global dynamic noprefixroute eth1  
        valid_lft 1747sec preferred_lft 1747sec  
    inet6 fe80::b4e0:7eee:eaab:9500/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

●Burp Suite(HTTP History)



The screenshot shows the Burp Suite interface with the 'HTTP history' tab selected. The table below represents the data shown in the history list.

#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes
1	http://update.googleapis.com	POST	/service/update2/json?cup2key=16:...	✓		200	13604	JSON			
2	http://edgedl.me.gvt1.com	GET	/edgedl/diffgen-puffin/ninodabcej...			200	76481	app			
3	http://update.googleapis.com	POST	/service/update2/json	✓		200	761	JSON			
4	http://edgedl.me.gvt1.com	GET	/edgedl/release2/chrome_compon...			200	5469	app	crx3		
5	http://update.googleapis.com	POST	/service/update2/json	✓		200	745	JSON			
6	http://edgedl.me.gvt1.com	GET	/edgedl/diffgen-puffin/bjbcblmdcn...			200	4421197	app			
7	http://update.googleapis.com	POST	/service/update2/json	✓		200	761	JSON			
8	http://edgedl.me.gvt1.com	GET	/edgedl/release2/chrome_compon...			200	5248	app	crx3		
9	http://update.googleapis.com	POST	/service/update2/json	✓		200	745	JSON			
10	http://edgedl.me.gvt1.com	GET	/edgedl/release2/chrome_compon...			200	17355	app	crx3		
11	http://update.googleapis.com	POST	/service/update2/json	✓		200	745	JSON			
12	http://localhost:3000	GET	/			200	75471	HTML		OWASP Juice Shop	

●Wireshark(HTTP Capture)

Capturing from Loopback: lo

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
28	24.583750308	127.0.0.1	127.0.0.1	HTTP	407	GET / HTTP/1.1
33	24.587929239	127.0.0.1	127.0.0.1	HTTP	407	GET / HTTP/1.1
37	24.932393416	127.0.0.1	127.0.0.1	HTTP	282	HTTP/1.1 200 OK
42	24.943719192	127.0.0.1	127.0.0.1	HTTP	282	HTTP/1.1 200 OK
73	40.818485700	127.0.0.1	127.0.0.1	HTTP	613	GET / HTTP/1.1
75	40.822943723	127.0.0.1	127.0.0.1	HTTP	459	HTTP/1.1 200 OK
77	41.193504562	127.0.0.1	127.0.0.1	HTTP	549	GET / HTTP/1.1
81	41.199690011	127.0.0.1	127.0.0.1	HTTP	550	GET / HTTP/1.1
83	41.200311838	127.0.0.1	127.0.0.1	HTTP	457	HTTP/1.1 200 OK
94	41.205302160	127.0.0.1	127.0.0.1	HTTP	549	GET / HTTP/1.1
96	41.205421245	127.0.0.1	127.0.0.1	HTTP	551	GET / HTTP/1.1

Frame 28: Packet, 407 bytes on wire (32000000) (capture length 407)

- Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 01:89:2b:6b:40:00 (01:89:2b:6b:40:00)
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Src Port: 54980, Dst Port: 80
- Hypertext Transfer Protocol

Hypertext Transfer Protocol: Protocol Packets: 628 - Displayed: 104 (16.6%) Profile: Default

●IP Configuration

```
(kali㉿kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:74:cf:dd brd ff:ff:ff:ff:ff:ff  
    inet 192.168.60.128/24 brd 192.168.60.255 scope global dynamic noprefixroute eth0  
        valid_lft 1365sec preferred_lft 1365sec  
    inet6 fe80::3ab6:112d:4852:6538/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:74:cf:e7 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.191.128/24 brd 192.168.191.255 scope global dynamic noprefixroute eth1  
        valid_lft 1365sec preferred_lft 1365sec  
    inet6 fe80::b4e0:7eee:eaab:9500/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
4: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default  
    link/ether 42:73:11:fc:b5:74 brd ff:ff:ff:ff:ff:ff  
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::4073:11ff:fefc:b574/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
7: veth71ae697@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default  
    link/ether ee:d9:c9:03:35:8e brd ff:ff:ff:ff:ff:ff link-netnsid 0  
    inet6 fe80::ecd9:c9ff:fe03:358e/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever
```


Command Used

Purpose	Command
Update System	<code>sudo apt update && sudo apt upgrade -y</code>
Install Docker	<code>sudo apt install docker.io -y</code>
Run Juice Shop	<code>sudo docker run -d -p 3000:3000 --name juice-shop bkimminich/juice-shop</code>
Check Docker status	<code>sudo docker ps</code>
Nmap Scan	<code>Nmap -p 3000 127.0.0.1</code>
IP Config	<code>ip a</code>
Burp Suite	<code>burpsuite</code>
Wireshark	<code>wireshark</code>

Conclusion

- Successfully built a personal cybersecurity lab with:
 - ✓ Kali Linux attacker machine
 - ✓ Juice Shop vulnerable target
 - ✓ Connectivity verified via Nmap
 - ✓ Burp Suite intercept working
 - ✓ Wireshark HTTP capture working